

WEEK 1 (Praveen 2520030406)

1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```
UW PICO 5.09
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    char cmd[100];
    pid_t pid;

    printf("Enter Linux command: ");
    scanf("%99[^\n]", cmd);

    pid = fork();

    if (pid < 0)
    {
        printf("fork() failed\n");
        exit(1);
    }

    else if (pid == 0)
    {
        // Child process
        printf("Child Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        execl("/bin/sh", "sh", "-c", cmd, NULL);

        // This executes only if exec fails
        printf("exec() failed\n");
        exit(1);
    }

    else
    {
        // Parent process
        printf("Parent Process\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);

        wait(NULL);

        printf("Child process completed\n");
    }

    return 0;
}
```

```
[praveenbusala@Praveens-MacBook-Air-2 CLG % nano week1.c
[praveenbusala@Praveens-MacBook-Air-2 CLG % gcc week1.c
[praveenbusala@Praveens-MacBook-Air-2 CLG % ./a.out
Enter Linux command: ls
Parent Process
Parent PID: 24036
Child PID: 24038
Child Process
Child PID: 24038
Parent PID: 24036
a.out          arithmetic.c      file.txt          p2.c
addition.c     code              filename.c        p3.c
arithmetic     code.c           multiplication.c  process1.c
Child process completed
praveenbusala@Praveens-MacBook-Air-2 CLG % █
```